

# The Systems Theoretical Perspective

A Quiet Revolution in Science

SEAtS | 558040

# Intended Learning Outcomes

By the end of this lecture, you will be able to:

1. Explain reductionism and trace it to Descartes' Cartesian dualism and Newton's mechanistic universe.
2. Define emergence and use examples to explain why parts alone don't explain wholes.
3. Distinguish the systems theoretical perspective, systems theory, systems thinking, and systems philosophy.
4. Describe the "funnel model" of systems-theoretical theories and critique its limitations.
5. Evaluate the ethical implications of systems thinking, contrasting a holistic reading against a technocratic one.

# Agenda

- What Is System Theory?
- What Is Systems Thinking?
- The Evolution of Systems Thinking
- Two Ways to See a Tree
- Reductionism: The Clockwork Universe
- Emergence: Why Parts Don't Explain Wholes
- The Interdisciplinary Insight
- Terminology: One Umbrella, Three Terms
- The Funnel Model
- Ethical Implications
- Conclusion

# What Is System Theory?

## Definition

System theory is a transdisciplinary framework for understanding how complex entities function as integrated wholes — examining systems through their components, relationships, and emergent behaviours, rather than isolating individual parts.

Core principles:

- **Interdependence** — components rely on each other
- **Feedback loops** — systems respond to their own outputs
- **Hierarchical structure** — systems nest within systems
- **Equifinality** — multiple pathways can produce the same outcome

# Why System Theory Matters

## For Cyber-Physical Systems

Traditional analysis breaks problems into components; system theory reveals how changes in one domain propagate through feedback loops to affect others.

- Sensor failures cascade into control instability
- Code updates alter how systems respond to real-world conditions
- Security threats can destabilize physical infrastructure

System theory is not a luxury — it is necessary for safe, robust, and resilient design.

# What Is Systems Thinking?

## Definition

Systems thinking is the practice of using system theory as a lens to understand and solve complex problems — examining relationships, feedback loops, delays, and emergent behaviours within the whole, rather than breaking problems into isolated parts.

Key practices:

- Mapping system structure and boundaries
- Identifying reinforcing and balancing feedback loops
- Recognizing non-linear cause-effect relationships
- Tracing how changes propagate across the system over time

# Why Systems Thinking Matters

## In Practice

Systems thinking resists reductionism — the false assumption that understanding parts means understanding wholes. It is essential wherever local optimization can create system-level harm: cyber-physical systems, software-driven organizations, policy design.

It prevents unintended consequences and reveals leverage points for effective intervention.

## Example: Social Media Feedback Loops

- User engages with political content
- Algorithm learns and recommends more of it
- More engagement trains the algorithm further
- Feed narrows to increasingly extreme content

No engineer programmed the polarization — it **emerged** from the feedback loop between algorithm and user behaviour.

# The Evolution of Systems Thinking

Four eras trace how systems thinking moved from biology, through modelling practice, across disciplines, to today's digital and AI-driven systems.

## 1930s–1950s: Origins in Systems Theory

Ludwig von Bertalanffy, a biologist, developed General Systems Theory in response to reductionism.

He observed that organisms, ecosystems, and organizations all exhibited common patterns — feedback loops, self-regulation, emergence — that mathematics and physics alone couldn't explain.

His *General System Theory* (1968) provided the foundational framework.

## 1960s–1970s: Systems Thinking Becomes Practice

System theory remained academic until practitioners applied it to real problems:

- **Jay Forrester** (MIT) developed *System Dynamics* to model complex systems with feedback loops and delays — applied to urban planning and resource management
- **The Club of Rome** used system dynamics to model global resource limits (*The Limits to Growth*, 1972)
- **Peter Senge** brought systems thinking to organizational management (*The Fifth Discipline*, 1990), emphasizing feedback loops in business

## 1980s–1990s: Expansion Across Disciplines

Systems thinking spread beyond academia:

- **Software engineering** — Donald Knuth, complexity theory; recognition that large systems behave unpredictably
- **Ecology** — Barry Commoner's "Laws of Ecology" apply systems thinking to environmental problems
- **Policy and governance** — Horst Rittel introduced "wicked problems": complex, interconnected problems that defy linear solutions

## 2000s–Present: Digital Age and Emergence

- **Network effects** — Internet, social media, and digital platforms revealed feedback loops at scale; algorithms and human behaviour became coupled systems
- **Complexity science** — emergence, agent-based modeling, power laws in networks
- **Cyber-physical systems** — IoT, autonomous vehicles, smart infrastructure: physical and computational systems tightly coupled
- **AI and systems thinking** — AI systems create feedback loops with user behaviour, producing emergent harms (filter bubbles, polarization, adversarial dynamics)

# From Theory to Practice

Systems thinking evolved from an academic framework into a practical methodology. Early work identified patterns; modern work applies them to messy problems where reductionism fails: climate change, pandemic response, platform governance, autonomous systems safety.

This history shows that cyber-physical systems, software platforms, and organizations are not new problems — they follow patterns recognized since the 1960s.

# Two Ways to See a Tree

A tree cannot be explained from itself alone:

- The sun supplies its energy
- A fungal network beneath the roots supplies nutrients — without it, the tree can't survive
- Reproduction depends on wind or animals

Reality is made of causal connections woven **horizontally** (between things) and **vertically** (across levels — atoms → cells → trees → forests). A tree can't be traced back to a single cause and a single effect.

# Reductionism: The Clockwork Universe

René Descartes (17th century) split the world into two substances:

| Substance           | Domain                                        |
|---------------------|-----------------------------------------------|
| <i>Res cogitans</i> | Mind — thoughts, feeling, consciousness       |
| <i>Res extensa</i>  | Matter — the outer, measurable physical world |

This strict separation is **Cartesian dualism**.

## Consequence

If matter has no soul or inner perspective, it can be taken apart, measured, and analysed without accounting for anything immaterial — "the whole is the sum of its parts."

**Newton** then modelled the universe as one giant, precise, predictable mechanism: know the laws, calculate any state, past or future.

# Reductionism: The Method

## Definition

To understand a complex system, break it into parts, understand the parts, and you understand the whole.

Body → organs → cells → molecules → atoms → quarks and electrons.

## Track Record

Decoding the human genome, particle accelerators (CERN), skyscrapers — reductionism is one of the most successful methods humanity has developed.

# Emergence: Why Parts Don't Explain Wholes

## The Neuron

A single neuron can be fully described (chemical reactions, electrical signalling). But how do billions of neurons give rise to a thought, a memory, or the feeling of hearing music?

## The Hurricane

A hurricane is only air and water molecules swirling in disorder — no single molecule "knows" about the storm. Yet their interplay produces a stable vortex holding its shape for days across thousands of kilometres.

**Emergence** — the order exists only in the relationships between parts, not in any one part. It is the first great riddle reductionism cannot solve.



# Turn and Talk

Can you think of another everyday example of emergence — a whole with a property that none of its parts have on their own?

**4:00** minutes

# The Interdisciplinary Insight

Scientific disciplines can't be thought of independently, because everything is connected, and the *same* phenomena show up across completely different disciplines.

"It is inculcated by our great universities [that] there's such a thing as psychology which is different from sociology... and that the world is made of separable items of knowledge... The world is not like that at all."

The body, the brain, human society, and an ant colony all self-organize — the same underlying phenomenon, studied by separate disciplines:

| Phenomenon                | Discipline |
|---------------------------|------------|
| The body / the ant colony | Biology    |
| The brain                 | Neurology  |
| The individual            | Psychology |
| Society                   | Sociology  |
| Markets                   | Economics  |

# Terminology: One Umbrella, Three Terms

- **Systems theoretical perspective** (*umbrella*) — the world as one connected system; attention on interactions between its parts.
  - **Systems theory** — the *hard science* layer: research, measurement, mathematical description. Numerous, sometimes contradictory theories.
  - **Systems thinking** — the *craft*: applying systems theory to concrete areas (psychology, health, management, personal life).
  - **Systems philosophy** ( $\approx$  *holism*) — metaphysical and ethical questions: meaning, ethics, the role of consciousness.
    - Whitehead — process philosophy: reality as process and relations, not fixed substance.
    - Hegel — a whole that unfolds through contradiction and change.

# The Funnel Model

Explicitly a simplification — view critically. Included as a useful overview, not a rigorous claim.

| Level         | Status                           | Theories / Thinkers                                                          |
|---------------|----------------------------------|------------------------------------------------------------------------------|
| Funnel closed | Fully reductionist               | Reductionism, atomism, classical physics                                     |
| Funnel opens  | Widely accepted, rigorous        | Dissipative structures (Nobel Prize); chaos theory; game theory; cybernetics |
| Rising        | Discussed, not fully accepted    | Integrated Information Theory; panpsychism; process philosophy (Whitehead)   |
| Upper         | Public resonance, not scientific | Fritjof Capra — physics and Taoism                                           |
| Lower         | Pre-scientific                   | Mystical and spiritual traditions                                            |

# The Funnel Model: The Lower Tiers

"The Cartesian–Newtonian worldview is embodied in our social institutions... What we need is a holistic or ecological worldview which takes into account the fundamental interdependence of all phenomena... and the world as a process rather than as a rigid structure."

— *Fritjof Capra*

The theories at the Lower of the funnel are read as early, symbolic attempts to grasp the universe as one connected, self-organizing system — to be handled critically, not adopted wholesale.



# Ethical Implications: Two Readings

## Holistic Reading

- We are part of the system — grounds an obligation to protect our environment
- Extends to social equality: a system works better when all its parts thrive
- We shouldn't presume to fully control nature — massive intervention risks an uncalculable chaotic tipping point
- Intervene less forcefully, trust self-organization, question power structures

## Technocratic Reading

- Extends from cybernetics: treat humans and environment as a machine to be steered
- Under this logic, even totalitarian control starts to look permissible
- Named examples: Peter Thiel (Palantir — data surveillance infrastructure); World Economic Forum (top-down, expert/data-steered society)
- Labelled "**technofeudalism**": digital surveillance and technical control of nature

# Ethical Implications: The Counter-Argument

Chaos theory and self-organization cut against the technocratic reading: turning all the knobs at once and trying to control people from above leads to chaos.

**A complex system cannot be steered like a machine.**

# Conclusion

The systems theoretical revolution is quiet — no manifesto, no single founder — but it may change how we see the world more profoundly than louder revolutions before it.

- Reductionism explains parts; it doesn't explain what emerges between them
- The same phenomena recur across disciplines that treat them as separate
- The perspective carries ethical weight both ways: toward humility, and toward control

Before pushing further into AI and other technologies, we should first work on understanding the world — and our place in it.

# Thank you

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